VaaniEHR — Research Dossier, SRS Inputs and PRD (Build-vs-Reuse Edition)

Bottom line: VaaniEHR can be built in about 300 hours only if you reuse almost everything except the parts that make it different. Those parts are the guideline-grounded question engine (M3), the safety-rule layer (M4) and the Gujarati benchmark. For the rest: use Sarvam Saaras (API) as the main ASR and AI4Bharat IndicConformer/IndicWhisper (MIT) as the open baseline, the NRCeS FHIR IG v6.5.0 with the HAPI/HL7 validator for M6, ABDM V3 sandbox APIs for ABHA (M1), the Meta WhatsApp Cloud API with a Gujarati ("gu") template for M7, and NLEM 2022 plus an NRCeS SNOMED CT affiliate licence for M1 (medicine master). Apply for the accounts and licences now, because sandbox and licence lead times are your biggest schedule risk.

TL;DR

- **Reuse the commodity layers; build only the differentiators.** ASR (Sarvam Saaras V4 at ₹30/hour, or IndicConformer MIT), FHIR (NRCeS IG 6.5.0 + HAPI), ABDM (V3 sandbox + Bahmni India/Nirmitee references), messaging (WhatsApp Cloud API) and evaluation (jiwer, Eka KARMA/medical ASR dataset) all exist. The STW-grounded question engine, the safety rules and a Gujarati acted-consultation benchmark do not exist anywhere public. That gap is your contribution.
- **Licences are the hidden trap.** DDInter (drug interactions) is CC BY-NC-SA 4.0. bioregistry Eka's Parrotlet-e is CC-BY-SA-4.0. SNOMED CT needs an NRCeS affiliate licence (free in India). nrces Scraped Indian brand datasets on Kaggle have unclear provenance. All are fine for a non-commercial research prototype, but each must be recorded in the SRS and in an SBOM, which the CDSCO 2026 MDSW guidance now expects. lexology
- **Regulatory framing is settled enough to write the SRS.** The CDSCO Guidance on Medical Device Software (CDSCO/MD/GD/MDSW/01/2026, 21 July 2026) is interpretive guidance under MDR 2017 and requires an intended-use statement, a QMS-style lifecycle and an SBOM. natlawreview + 2 Under the DPDP Rules 2025 (notified 13 November 2025), the Consent Manager framework starts 13 November 2026 and the substantive duties start 13 May 2027. ispectratechnologies Position VaaniEHR as a research prototype that only "informs" clinical management, with mandatory doctor sign-off. asiaactual

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Key Findings

1. **ASR is solved enough to reuse, but Gujarati medical accuracy is not.** On the official Vistaar table, IndicWhisper's Gujarati WER is 17.8% (Kathbath), 23.5% (FLEURS) and 33.2% (MUCS), averaging 22.8%. github Those are general-domain numbers. No published Gujarati *medical* WER was found. Eka's medical scribe lists Indian English, Hindi, Marathi, Kannada and Telugu but not Gujarati, and Eka's open Parrotlet-a-en-5b model is English-only. eka eka A Gujarati medical ASR benchmark would therefore be a genuine contribution.
2. **Sarvam is the pragmatic primary ASR.** It costs ₹30/hour (₹45 with diarization), is billed per second and comes with ₹100 free credit. smallest sarvam Saaras V4 was released on 25 September 2026; MarkTechPost reports it is served only through Sarvam's API (`model="saaras:v4"`) and that "weights are not public". MarkTechPost's comparison table lists keyterm biasing of "up to 50 terms" (useful for medicine names) and a reported 16.03% WER on IndicContextEval with keyword prompting. V4 also offers a code-mix output mode, WebSocket streaming with a first token under 150 ms, REST for clips up to 30 s, and batch jobs up to 2 h. At about 10 minutes per consultation, 300 acted consultations come to about 50 audio hours: roughly ₹1,500 at base rate or ₹2,250 with diarization.
3. **ABDM V3 is the current API generation, but sandbox behaviour is uneven.** Nirmitee's ABDM v3 migration guide states that "NHA's official ABDM Wrapper has migrated to V3, and v0.5 gateway endpoints increasingly return 403 errors". In the Nirmitee-tech ABDM V3 Postman collection README on GitHub, the HIP-initiated endpoints `/v3/token/generate-token`, `/hip/v3/link/carecontext` and `/hip/v3/link/context/notify` are marked "403", while the user-initiated linking callbacks are marked "Working". This supports keeping M2 (care-context linking) as a stretch goal.
4. **The NRCeS FHIR IG 6.5.0 (package ndhm.in, FHIR 4.0.1, generated 2025-05-08) is the target.** nrces + 3 It includes OPConsultRecord, PrescriptionRecord, MedicationRequest, Patient, Practitioner and the other profiles you need. nrces nrces
5. **ICMR STWs are one-page PDFs, not machine-readable rules.** google icmr You will have to hand-encode about 5 STWs into decision trees. That is the core intellectual work of M3, and it is what makes your rules-only vs LLM-only vs hybrid comparison meaningful.
6. **No open Indian drug-interaction dataset with a permissive licence was found.** DDInter 2.0 is the best fit for M4. According to Tian et al. (Nucleic Acids Research 2025;53(D1):D1356–D1362, doi:10.1093/nar/gkae726), it covers 2,310 drugs with 302,516 drug-drug interaction records and 6,033 therapeutic duplication interactions. It is non-commercial.

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1. ASR / Speech (M2)

Resource	What it is	Link	Licence	Last activity	How VaaniEHR uses it
Sarvam Saaras V4 / Saarika (API)	Indic STT for 22 Indian languages + English, with transcribe/translate/verbatim/translit/codemix modes callmissed marktechpost	https://docs.sarvam.ai/api/getting-started/pricing	Commercial API	V4 announced Sep 2026	Primary streaming ASR f switched speech; keytern names
AI4Bharat IndicConformer-600M multilingual	Hybrid CTC+RNNT Conformer for 22 languages incl. Gujarati huggingface	https://huggingface.co/ai4bharat/indic-conformer-600m-multilingual	MIT (model card)	HF collection updated Mar 2	Open-source baseline an
IndicConformer repo (monolingual models + AI4Bharat NeMo)	Training/inference code, per-language models incl. gu github	https://github.com/AI4Bharat/IndicConformerASR	MIT	Active	Gujarati-only model for c
Quantized IndicConformer (ONNX int8)	Community CPU-friendly quantization huggingface	https://huggingface.co/atharva-again/indic-conformer-600m-quantized	Inherits upstream (verify)	—	GPU-free CPU inference
Vistaar / IndicWhisper	59 benchmarks, 12 languages, 10,700 h of training data; Whisper fine-tunes github	https://github.com/AI4Bharat/vistaar	MIT	Paper Interspeech 2023	Benchmark protocol and baselines
Eka Medical ASR Evaluation Dataset	3,900+ medical audio clips (entities, sentences, conversations) eka	https://huggingface.co/datasets/ekacare/eka-medical-asr-evaluation-dataset	MIT	Updated Jul 30, 2025 huggingface	Medicine-name accuracy
Eka KARMA-OpenMedEvalKit	Medical AI eval toolkit incl. semantic WER and entity WER digitalhealthnews	https://www.eka.care/services/introducing-karma-openmedevalkit-an-open-source-framework-for-medical-ai-evaluation	Open source (licence not confirmed on repo)	2025	Entity-WER metric desig
Eka public datasets collection	ASR, clinical note, IndicMTEB, protocol QnA datasets huggingface	https://huggingface.co/collections/ekacare/ekacare-medical-public-datasets	Per-dataset	Updated 2026	`indian_protocols_ba` for M3 testing
Parrotlet-e	Indic medical embedding model aligned to SNOMED CT/UMLS, 12 Indic languages + English huggingface	https://huggingface.co/ekacare/parrotlet-e	CC-BY-SA-4.0	Nov 2025	Map spoken Gujarati/Hin medicine mentions to SN
faster-whisper	CTranslate2 Whisper re-implementation github	https://github.com/SYSTRAN/faster-whisper	MIT	v1.2.1 (Oct 2025) github	CPU Whisper baseline
whisper.cpp	C/C++ Whisper inference	https://github.com/ggml-org/whisper.cpp	MIT	v1.9.3, Aug 2026 releasealert	Fully local fallback

Resource	What it is	Link	Licence	Last activity	How VaaniEHR uses it
Silero VAD	Voice activity detection	https://github.com/snakers4/silero-vad	MIT	Active (Sep 2026)	Chunk audio before ASR;
pyannote-audio	Speaker diarization toolkit	https://github.com/pyannote/pyannote-audio	MIT (code; pretrained pipelines may need HF terms acceptance — unverified)	4.0.7 (2026) github	Doctor vs patient turn sep
Bhashini / ULCA	Government Indic speech/translation APIs	Not verified in this pass (search "Bhashini ULCA API")	Gov terms	—	Free alternative ASR/tran
IndicVoices	Large Indic speech dataset	Not verified in this pass (cited in the arXiv 2512.10967 bibliography) arxiv	—	—	Possible fine-tuning data

Judgement: Use Sarvam for the product path and IndicConformer for the open baseline, and report both. Do not fine-tune anything. Budget roughly ₹2,500 for ASR.

2. Medical NLP, Extraction and LLM Tooling

Resource	What it is	Link	Licence	Last activity	Use in VaaniEHR	Recommendation
OpenScribe	Open-source AI scribe: Whisper (whisper.cpp) + Claude/Ollama notes	https://github.com/sammargolis/OpenScribe	MIT	Active	Reference pipeline (ingest → transcribe → LLM → note editor)	**Study / borrow architecture** github
FreeScribe (ClinicianFOCUS)	Whisper + local LLM SOAP scribe, consent-first workflow	https://github.com/ClinicianFOCUS/FreeScribe	Not confirmed	Active	Consent-before-record UX pattern	Study
scribeHC	Expo + Next.js + FastAPI scribe with OpenAI APIs	https://github.com/trevorpfiz/scribeHC	Not confirmed	—	FastAPI audio-service structure	Study
Awesome-Medical-Scribe-AI	Curated list (updated Aug 2026)	https://github.com/ishandutta2007/Awesome-Medical-Scribe-AI	List	Aug 2026	Competitor/related-work table for the report	Use for literature review github
medspaCy	Clinical NLP incl. ConText negation/assertion github	https://github.com/medspacy/medspacy	MIT	Jun 2026 (v1.3.1) pypi github	Negation baseline on English/translated text	Baseline only (English-centric)
Outlines	Constrained/structured generation hellogithub	https://github.com/dottxt-ai/outlines	Apache-2.0	Sep 2026	JSON-schema-constrained extraction with local models	Pick if self-hosting an LLM
Instructor	Pydantic-validated structured outputs over LLM APIs hellogithub	https://github.com/567-labs/instructor	MIT	1.15.1 (~May 2026) ecosyste	Extract `{symptom, onset, duration, negated, severity}` objects with retries	**Primary pick** with an API LLM
pgvector	Vector search inside PostgreSQL	https://github.com/pgvector/pgvector	PostgreSQL License atlan	v0.8.6, Jul 2026 releasealert	RAG over encoded STW text and NLEM monographs; also settles Postgres vs MySQL	**Pick** (removes the need for Chroma)
Sarvam LLMs (e.g., Sarvam-M 24B)	Indic-tuned LLMs	Pricing page above	API/open weights (verify)	2025–26	Gujarati understanding / summarisation	Evaluate; Gujarati quality unverified
Llama / Qwen / Gemma	Open LLMs	Not verified in this pass	Varies (Llama community licence is not OSI)	—	Local LLM-only baseline	GPU needed for useful sizes; prefer an API

Judgement: Use one API LLM with Instructor for extraction and a schema with explicit `negated` and `uncertain` flags. The hallucination metric is any extracted fact not grounded in a transcript span, so store character offsets.

3. Guided-Question Engine and Decision Support (M3)

Resource	What it is	Link	Licence	Last activity	Use	Recommenda
ICMR Standard Treatment Workflows	One-page PDF workflows by specialty; mobile app	https://www.icmr.gov.in/standard-treatment-workflows-stws	Government publication (reuse terms not stated; cite and attribute)	App updated Jan 2026	Source of truth for questions, red flags and first-line drugs	**Core source** i apple
STW methodology paper (PMC)	Describes STW development; 122 conditions in 23 disciplines across Vols 1–3	https://pmc.ncbi.nlm.nih.gov/articles/PMC10472454/	Open access	2023	Cite in the SRS for clinical validity	Cite
ICMR STW example (Cataract)	Shows format: symptoms, signs, ICD-10 code, levels of care	https://www.icmr.gov.in/icmrobject/uploads/STWs/1726569103_cataract.pdf	Gov	2024	Template for your JSON encoding	Reference icm
MediQ	Interactive question-asking benchmark (Patient + Expert systems)	https://github.com/stellalisy/MediQ	See repo	2024–25	Borrow the abstain/ask/answer loop and the "number of questions" metric	**Reuse the evaluation de** github arxiv
MediQ paper	Li et al. (University of Washington), NeurIPS 2024: "directly prompting state-of-the-art LLMs to ask questions degrades performance", while adding abstention improves diagnostic accuracy by 22.3%	https://arxiv.org/abs/2406.00922	arXiv	2024	Justifies the hybrid design	Cite
json-rules-engine	JSON-defined rules for Node npmjs	https://github.com/CacheControl/json-rules-engine	ISC	7.3.2, Feb 2026 github	Encode STW decision trees and safety rules as versioned JSON in the Node backend	**Primary pick**

Resource	What it is	Link	Licence	Last activity	Use	Recommenda
Bahmni openmrs-module-cdss	FHIR-based CDSS integration: drug–diagnosis, drug–drug, dosage	https://github.com/Bahmni/openmrs-module-cdss	See repo	—	Design reference for CDS Hooks-style safety calls	Study
Durable Rules, CLIPS, Drools, DDXPlus, MedDialog, ReMeDi, Symptom2Disease, eSanjeevani CDSS (OLDCART), AMIE, IndicMedDialog	Alternatives / datasets	Not verified in this pass	—	—	Related work	Mark as unverified in t report

Design recommendation: Represent each STW as a graph of nodes (question, answer type, SNOMED code), edges (conditions) and red-flag terminal nodes. The LLM only (a) maps utterances to answer slots and (b) phrases the next question in Gujarati. The rules choose *which* question to ask. Track top-1/top-3 accuracy and the number of questions asked, following the MediQ approach.

4. Medicine Database and Safety (M1, M4)

Resource	What it is	Link	Licence	L a
NLEM 2022 (CDSCO)	384 medicines in 27 therapeutic categories; released 13 Sep 2022	https://cdsco.gov.in/opencms/opencms/en/consumer/Essential-Medicines/	Gov	2
NLEM 2022 Committee report	Additions/deletions; 440 line items counting repeats	https://cdsco.gov.in/opencms/resources/UploadCDSCOWeb/2018/UploadPublic_NoticesFiles/reportnlem2022.pdf	Gov	2
NRCeS SNOMED CT + Common Drug Codes for India (CDCI)	National drug extension; India refsets; Hindi patient-instruction extension (490 concepts)	https://www.nrces.in/standards/snomed-ct and https://nrces.in/services/national-releases	Affiliate licence (free in India)	–
DDInter 2.0	302,516 DDIs, 2,310 drugs, therapeutic duplications, drug–disease interactions	https://ddinter2.scbdd.com	CC BY-NC-SA 4.0	P 2 (2
A-Z Medicine Dataset of India (Kaggle)	~250K brands, composition, price (Nov 2022)	https://www.kaggle.com/datasets/shudhanshusingh/az-medicine-dataset-of-india	Unclear (scraped)	2
Extensive A-Z Medicines Dataset	Merged brand/substitute/side-effect set	https://www.kaggle.com/datasets/riturajsingh2004/extensive-a-z-medicines-dataset-of-india	Labelled MIT (provenance still scraped)	–
junioralive/Indian-Medicine-Dataset	JSON/CSV of Indian brands	https://github.com/junioralive/Indian-Medicine-Dataset	See repo	–
RxNorm, WHO ATC/DDD, openFDA, DrugBank open, Jan Aushadhi list	Reference vocabularies	Not verified in this pass	Varies (WHO ATC has use restrictions; DrugBank open is CC BY-NC)	–

Max-dose / allergy data: No open, licence-clean Indian dose-range dataset was found. Hand-curate max daily doses for the ~40–60 NLEM drugs used across your 5 conditions, sourced from the National Formulary of India or the product label, and store the source per row. Allergy cross-reactivity (e.g., penicillin–cephalosporin) should be a small hand-written rule table.

5. ABDM and FHIR (M6)

Resource	What it is	Link	Licence	Last activit
NRCeS FHIR IG for ABDM v6.5.0	Profiles, value sets, examples	https://nrces.in/ndhm/fhir/r4/index.html	Gov	Gener 2025-0 08 nrc
OPConsultRecord profile	Composition profile for OP consult nrces	https://www.nrces.in/ndhm/fhir/r4/StructureDefinition-OPConsultRecord.html	Gov	6.5.0
Profiles index (PrescriptionRecord, MedicationRequest, Patient, Practitioner, Condition, Observation…)	—	https://www.nrces.in/ndhm/fhir/r4/profiles.html	Gov	6.5.0
NRCeS "Implementation Guide for Adoption of FHIR in ABDM and NHCX"	Pitfalls in bundle creation, validation steps, HAPI usage nrces	https://www.nrces.in/download/files/pdf/Implementation_Guide_for_Adoption_of_FHIR_in_ABDM_and_NHCX.pdf	Gov	—
HL7 FHIR validator (validator_cli)	Official validator jar	https://github.com/hapifhir/org.hl7.fhir.core	Apache-2.0 (unverified)	Freque releas
HAPI FHIR JPA server starter	Dockerised FHIR server	https://github.com/hapifhir/hapi-fhir-jpaserver-starter	Apache-2.0	Active
Medplum	TypeScript FHIR platform + React components github	https://github.com/medplum/medplum	Apache-2.0	Very active
fhir.resources (Python)	Pydantic FHIR models	https://github.com/nazrulworld/fhir.resources	BSD-3 (unverified)	Unver
ABDM V3 Postman collection (Nirmitee)	M1/M2/M3 V3 endpoints, sandbox + prod envs github	https://github.com/Nirmitee-tech/abdm-v3-postman-collection	MIT	2026
ABDM sandbox docs mirror	V3 sequence diagrams, server URLs github	https://kiranma72.github.io/abdm-docs/2-milestone1/verify-abha-address/health-facility-qr-scan/index.html	Community mirror	—

Resource	What it is	Link	Licence	Last activit
CoronaSafe ABDM implementer guide	Sandbox integration and exit process, M1/M2 end-state coronasafe	https://docs.coronasafe.network/abdm-documentation/implementers-guide/abdm-sandbox-integration-and-exit-process	Open docs	—
Bahmni India Distro	ABDM-certified HIP/HIU: hip-service, openmrs-module-hip, ABHA-Verification, docker package	https://github.com/bahmniindiadistro	MIT (modules)	Jan–F 2025
Bahmni HIP architecture page	—	https://bahmni.atlassian.net/wiki/spaces/BAH/pages/2901114904	—	—
Nirmitee guides (vendor blogs)	V3 step-by-step, HIP M2 callbacks nirmitee nirmitee	https://nirmitee.io/blog/step-by-step-guide-for-abdm-integration/	Blog	2026

Key ABDM V3 facts for the SRS (from practitioner sources; confirm on the official sandbox portal):

- ABHA sandbox base: `https://abhasbx.abdm.gov.in/abha/api`. Gateway sandbox: `https://dev.abdm.gov.in/hiecm/api`. github medium
- M1 endpoints: `/v3/enrollment/request/otp`, `/v3/profile/login/request/otp`, `/v3/profile/account/abha/search`. medium
- M2 endpoints: link token via `/api/hiecm/v3/token/generate-token` (delivered on callback), then `/api/hiecm/hip/v3/link/carecontext`, one care context per call. nirmitee
- Aadhaar and OTP values are RSA-encrypted. nirmitee
- The ABHA card download fails with 406 if an `Accept` header is sent. nirmitee

6. WhatsApp and Gujarati Output (M7)

Resource	What it is	Link	Licence	Use	Recommendation
WhatsApp message templates docs	Templates must be approved; Gujarati language code `gu` supported	https://developers.facebook.com/docs/whatsapp/api/messages/message-templates/	Meta terms	Gujarati "visit summary" utility template	**Core**
fbsamples/whatsapp-api-examples	Official Node samples (webhooks, templates, media)	https://github.com/fbsamples/whatsapp-api-examples	Meta Platform Policy	Copy webhook + template sender	**Reuse**
meta-cloud-api (TypeScript SDK)	Maintained SDK; Meta's own Node SDK archived since June 2023	https://github.com/froggy1014/meta-cloud-api	See repo	Typed client in Node	Pick github
IndicTrans2	En↔22 Indic translation; model checkpoints MIT	https://github.com/ai4bharat/IndicTrans2	MIT (checkpoints), CC0/CC-BY-4.0 data	English summary → Gujarati (guj_Gujr)	Pick if not using Sarvam translate github arxiv
Twilio, Gupshup, 360dialog	BSP alternatives	Not verified in this pass	Commercial	Fallback if Meta verification stalls	Alternative

Template design: Keep the template generic, with variables (e.g., "નમસ્તે {{1}}, તમારી મુલાકાતનો સારાંશ: {{2}}"), and send the doctor-approved Gujarati text as a variable. github gurusup Better still, send a PDF document link, because templates cannot carry free-form clinical text reliably. Never send before sign-off.

7. Frontend Building Blocks

Resource	Use	Status
Medplum React components (https://github.com/medplum/medplum)	Study FHIR-aware forms and resource tables	Verified Apache-2.0 github
OpenScribe note editor (https://github.com/sammargolis/OpenScribe)	Transcript → note editing UX	Verified MIT github
RecordRTC / react-media-recorder / wavesurfer.js / @react-pdf/renderer / shadcn-ui / TanStack Table	Recorder, waveform, PDF prescription, components	**Not verified in this pass.** The browser MediaRecorder + AudioWorklet → WebSocket (Socket.IO already in ClinicOS) is enough without extra libraries

8. Evaluation Tooling

Resource	Link	Licence	Use
jiwer	https://github.com/jitsi/jiwer (unverified)	Apache-2.0 (unverified)	WER/CER
Eka KARMA + medical ASR dataset	Links above	MIT dataset	Entity WER, semantic WER methodology digitalhealthnews
MediQ	https://github.com/stellalisy/MediQ	—	Question-engine evaluation loop github
HL7 validator in CI	https://github.com/hapifhir/org.hl7.fhir.core	—	FHIR validity = % bundles with 0 errors
"ASR Under the Stethoscope" (arXiv 2512.10967)	https://arxiv.org/pdf/2512.10967	arXiv	Related work on clinical ASR bias across Indian languages arxiv
Agricultural Indic ASR benchmark (arXiv 2602.03868)	https://arxiv.org/pdf/2602.03868	arXiv	Shows diarization-aware "best speaker" WER methodology arxiv

(a) Build vs Reuse Bill of Materials

Architecture component	Decision	Concrete resource
Consultation UI (React 18 + Tailwind)	Build	ClinicOS shell; OpenScribe UX as reference
Audio capture + streaming	Build (thin)	MediaRecorder → Socket.IO → Python ASR service
VAD	Reuse	Silero VAD (MIT)
ASR (primary)	Reuse (API)	Sarvam Saaras V4, codemix mode + keyterms from NLEM
ASR (baseline)	Reuse	IndicConformer-600M (MIT) / IndicWhisper (Vistaar, MIT)
Diarization	Reuse	Sarvam ₹45/h tier (or pyannote)
Fact extraction	Build on reuse	API LLM + Instructor (MIT) + own JSON schema
Entity linking to SNOMED	Reuse	Parrotlet-e (CC-BY-SA-4.0) + NRCeS SNOMED CT release
Question engine (M3)	**Build**	ICMR STWs hand-encoded → json-rules-engine (ISC)
RAG over STWs	Reuse	pgvector in PostgreSQL 16
Medicine master	Build on reuse	NLEM 2022 CSV + CDCI SNOMED codes + optional Kaggle brands
Safety checks (M4)	**Build**	Own rules + DDInter 2.0 (CC BY-NC-SA) + curated max-dose table
Sign-off + audit (M5)	Build	ClinicOS `audit_log`, append-only, hash-chained
FHIR bundle builder (M6)	Build on reuse	NRCeS IG 6.5.0 examples; Medplum fhirtypes or fhir.resources
FHIR validation	Reuse	HL7 validator_cli in GitHub Actions
ABHA verification	Reuse	ABDM V3 sandbox + Nirmitee Postman collection
Care-context linking (stretch)	Reuse/reference	Bahmni India hip-service design
WhatsApp summary (M7)	Reuse	WhatsApp Cloud API + fbsamples + IndicTrans2/Sarvam translate
ClinicOS link (M8)	Build	One REST endpoint + webhook
Evaluation	Reuse	jiwer, KARMA metrics, MediQ loop
Deployment	Reuse	Docker Compose: `web`, `api-node`, `asr-py` (FastAPI), `postgres+pgvector`; Redis optional

(b) Accounts, Credentials and Approvals — Apply This Week

1. **ABDM Sandbox client ID/secret** and HIP ID (M1, then M2). Unknown lead time; it is the biggest risk.
2. **NRCeS SNOMED CT Affiliate Licence** via MLDS (https://mlds.ihtsdotools.org/#/landing/IN). The download link arrives in 4–5 business days after acceptance. nrces
3. **Meta Business account + WhatsApp Cloud API** test number; submit the Gujarati utility template early.
4. **Sarvam API key** (₹100 free credit; top up about ₹2,500).
5. **Hugging Face account** (accept any gated-model terms for IndicConformer/pyannote).
6. **Bhashini/ULCA account** (optional backup).
7. **Institutional Ethics Committee / faculty approval** for recording acted consultations, plus written consent and an audio-release form from each actor.
8. **LLM API key** (with a "no training on data" setting).

(c) Facts and Standards for SRS Non-Functional / Regulatory Sections

- **Intended use (CDSCO/MD/GD/MDSW/01/2026, 21 July 2026):** The guidance says intended use should state the medical purpose, the disease and its criticality, the patient population, users, use environment, contraindications, software function and platform. natlawreview The CDSCO matrix grades software that "informs clinical management" in a non-serious situation as Class A, and in a serious situation as Class B. Treatment or diagnosis software reaches Class C/D. asiaactual VaaniEHR must stay in the "inform" bucket, with mandatory clinician sign-off. The guidance also expects a lifecycle QMS (design, V&V, configuration, change, release, maintenance) and an **SBOM**, and introduces an optional Algorithm Change Protocol. lexology + 2 It is guidance under MDR 2017, not a new law. lexology + 2 Final PDF: https://cdsco.gov.in/opencms/export/sites/CDSCO_WEB/Pdf-documents/Guidance-document-on-Medical-Device-Software-under-MDR-2017.pdf
- **DPDP:** The Rules were notified 13 Nov 2025. ispectratechnologies Rule 4 (Consent Managers) applies from 13 Nov 2026. Rules 3 and 5–16 (notice, consent, security safeguards, breach notification within 72 hours, data principal rights, retention) apply from 13 May 2027. privybyidfy + 2 SRS requirements: an itemised notice, separate consent for audio recording and for AI processing, withdrawal, erasure, a breach runbook, and Gujarati notice text.
- **Interoperability:** FHIR R4 4.0.1; NRCeS IG ndhm.in 6.5.0; nrces nirmitee SNOMED CT (NRCeS affiliate licence; end users need a sublicence); nrces ABHA number (14-digit) and ABHA address stored per patient. nirmitee
- **Security:** Treat the OWASP Top 10 for LLM Applications as a requirement source. The 2025 list is LLM01 Prompt Injection, LLM02 Sensitive Information Disclosure, LLM03 Supply Chain, LLM04 Data and Model Poisoning, LLM05 Improper Output Handling, LLM06 Excessive Agency, LLM07 System Prompt Leakage, LLM08 Vector and Embedding Weaknesses, LLM09 Misinformation, LLM10 Unbounded Consumption. owasp OWASP's GitHub states that the "OWASP GenAI LLM Top 10 2026" was published on 4 August 2026. That edition re-orders these items and renames System Prompt Leakage as "Hidden Context Exposure"; cite the 2026 list as current. OWASP ASVS Level 2 for the web app (unverified version). JWT + RBAC, TLS, encryption at rest for audio.
- **Lifecycle standards:** ISO/IEC/IEEE 29148:2018 (SRS structure), IEC 62304 (software lifecycle, safety class), IEC 82304-1 (health software) and WCAG 2.1 AA. None was re-verified in this pass; cite them by name.
- **NMC generic-prescribing rule status:** not verified in this pass. State "generic-first by design" without asserting the current legal status.

(d) Pitfalls

- **Non-commercial data:** DDInter, possibly WHO ATC, and DrugBank open are non-commercial. bioregistry If ClinicOS is ever commercialised, M4 needs a replacement source.
- **Gated or "MIT" models:** Verify access and licence at download time. One paper calls IndicConformer gated even though the model card says MIT. huggingface arxiv
- **Sandbox instability:** Some V3 HIP-initiated endpoints return 403 in community tests. github Do not put M2 on the critical path.
- **Postgres vs MySQL:** pgvector and JSONB argue for **PostgreSQL 16**. Settle this before writing migrations.
- **Prices and versions:** Saaras V4 is new (Sept 2026) and API-only, marktechpost so pin the model version used in evaluation.
- **Acted data ≠ real data:** Report this as a limitation. Use several Gujarati speakers and accents, plus background noise.
- **LLM-generated Gujarati summaries** must be doctor-approved text only. Log the exact message sent.

PRD — VaaniEHR (shareable with developers / AI builders)

1. Overview and Goals

Product: VaaniEHR, a web module that listens to a general-medicine OPD consultation in Gujarati/Hindi/English, asks guideline-based follow-up questions, suggests safety-checked NLEM medicines, and produces a doctor-signed, ABDM-ready FHIR record plus a Gujarati WhatsApp summary.
Positioning: A research prototype using acted/simulated data only. It is not for clinical use. **Goals:**

- (G1) Cut documentation time per visit.
- (G2) Have zero unsafe suggestions reach sign-off undetected.

- (G3) Make 100% of signed visits produce a valid NRCeS FHIR bundle.
- (G4) Publish a Gujarati acted-consultation benchmark.

2. Target Audience

- **Primary:** General-practice doctors in Gujarat-region OPDs (adult patients). They are bilingual or trilingual, time-poor and use a phone or laptop.
- **Secondary:** Clinic staff (registration, ABHA capture), patients (receive the Gujarati summary), and the clinic admin (templates, audit).
- **Research users:** Faculty and evaluators running the benchmarks.

3. Required Pages

1. Login (ClinicOS JWT)
2. Doctor dashboard (today's tokens, drafts awaiting sign-off)
3. Patient lookup/registration, with ABHA verify (sandbox)
4. Consent capture (recording + AI processing, Gujarati/English, withdraw)
5. **Consultation workspace:** live transcript, extracted facts, guided-question panel, red-flag banner
6. Diagnosis review (top-3 with STW citations)
7. **Prescription builder:** generic-first NLEM autocomplete, brand optional, templates, inline safety alerts
8. Review and sign-off (diff of AI vs doctor edits; sign = commit)
9. Visit record / FHIR viewer (bundle JSON + validation result)
10. Patient summary preview and WhatsApp send log
11. Medicine master and prescription-template admin
12. Rules/STW library admin (versioned decision trees)
13. Audit log viewer
14. Evaluation console (upload acted audio, run baselines, metrics dashboard)
15. Settings (API keys, language, intended-use statement / About page)

4. Core Features and Priority

ID	Feature	Priority
F1	Consent gate before any recording	P0
F2	Streaming code-switched ASR with transcript editing	P0
F3	Fact extraction with negation and source-span links	P0
F4	STW-based guided questions (5 conditions), red-flag stop	P0
F5	NLEM medicine DB with SNOMED codes, generic-first	P0
F6	Safety checks: allergy, max dose, duplicate class, DDI, red flag	P0
F7	Doctor sign-off; nothing persisted as final before sign-off; audit log	P0
F8	FHIR OPConsultRecord/Prescription bundle + validator	P0
F9	ABHA verification (sandbox, M1)	P1
F10	Gujarati WhatsApp summary after sign-off	P1
F11	Prescription templates	P1
F12	Evaluation console and baselines	P1
F13	ClinicOS single-API integration	P1
F14	HIP care-context linking (M2)	P2
F15	Prescription PDF	P2

5. User Journeys

- **Doctor consultation (happy path):** Pick a token → confirm consent → press Record → speak naturally → transcript and facts appear → the question panel suggests "Any blood in sputum?" (from the STW) → the doctor asks it, and the answer is captured → top-3 diagnoses with

citations appear → the doctor selects one → suggested NLEM drugs appear with green/amber/red checks → the doctor edits the doses → reviews the diff → signs → FHIR bundle validated → Gujarati summary sent.

- **Red-flag path:** A red flag is detected (e.g., chest pain + sweating) → a blocking banner shows "refer/urgent" → medicine suggestions are suppressed → the doctor acknowledges, and the acknowledgement is logged.
- **Staff registration:** Phone lookup → new patient → ABHA verify via OTP (sandbox) → ABHA number/address saved → token issued.
- **Patient:** Receives the WhatsApp template in Gujarati with the summary and medicine schedule.
- **Researcher:** Uploads acted audio + gold annotations → runs rules-only / LLM-only / hybrid → views metrics.

6. Success Metrics (targets for the acted benchmark; set as hypotheses)

- ASR: WER/CER on Gujarati code-switched speech, reported against the Vistaar Gujarati baselines (17.8–33.2% general domain); medicine-name accuracy ≥90% with keyterm biasing.
- Extraction: precision/recall ≥0.85, including negated findings; hallucination rate (ungrounded facts) ≤2%.
- Question engine: top-3 accuracy ≥85%, top-1 ≥65%; median questions ≤6. The hybrid should beat rules-only and LLM-only.
- Safety: 100% of seeded unsafe scenarios flagged; 0 unsafe items signed without an override reason.
- FHIR: 100% of signed visits validate with 0 errors against ndhm.in 6.5.0.
- Workflow: documentation time per visit lower than a manual-typing baseline; SUS score from the doctors who test it.

Caveats

- I could not open some sources directly, and verified some repos only through search snippets. Unverified items are marked in the tables: jiwer, fhir.resources and HL7 validator licences; RecordRTC and other UI libraries; Bhashini; RxNorm/ATC/openFDA; dialogue datasets; IEEE/IEC/WCAG texts; NMC rules.
- ABDM V3 endpoint details come from practitioner and vendor sources (Nirmitee, community docs), not the official portal. Confirm them after you get sandbox access.
- The metric targets above are proposed hypotheses, not published benchmarks.